



Computer Science Year 2

Algorithms & Data

Estimation, Regression, Classification

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🔥 Last time ...

- Bayesian methods
 - In Bayesian approach the unknown parameter is assumed to be a random variable;
 - They enable prior information about the parameters to be incorporated in the estimation procedure;
 - They do not need to be justified by any asymptotic approximation;
 - Bayesian techniques are based on modelling the uncertainty with respect to the parameter θ through a probability distribution.
- *Disadvantages:-*
 - A prior distribution must be specified. This presupposes more work and can be subjective
 - Except for some special cases of prior distributions (e.g. Gaussian, Cauchy, exponential, Laplacian), the derivation of the posterior distribution is cumbersome and requires numerical methods.

🔥 Bayesian estimators

- In general, a Bayesian estimator minimizes the conditional risk, which is the loss (cost function) averaged over the conditional (posterior) distribution of θ , given the observation (measurement) x :

$$\hat{\theta}(x) = \operatorname{argmin}_{\theta} \int C[\theta, \hat{\theta}(x)] p(\theta|x) d\theta$$

- Definition: *The Bayes risk R is the average cost $E[C(\varepsilon)]$ and measures the performance of a given estimator.*

$$R = E[C(\varepsilon)]$$

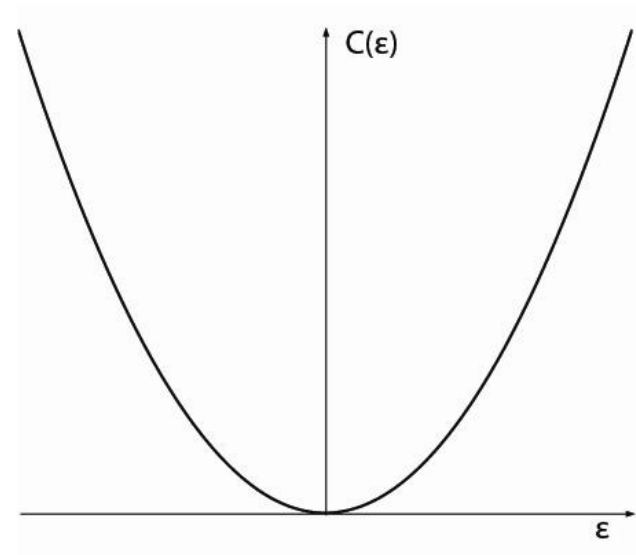
🔥 The Minimum Mean Square Error (MMSE) estimator

- Quadratic error cost function

$$C[\theta, \hat{\theta}(x)] = C(\varepsilon) = \varepsilon^2$$

- The corresponding optimal estimator is the *mean of the posterior PDF*

$$\hat{\theta} = \int \theta p(\theta|x) d\theta = E(\theta|x)$$



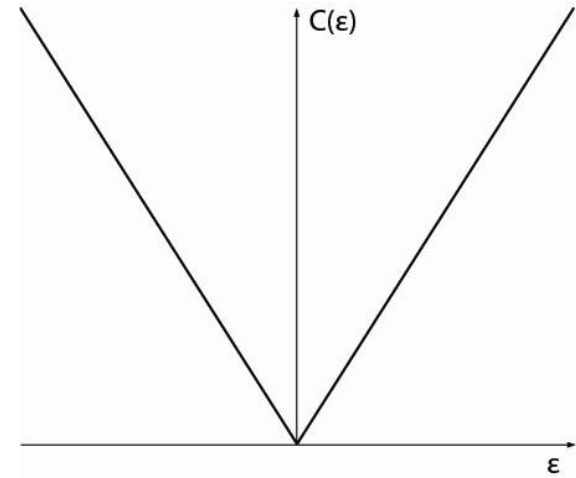
🔥 The Minimum Absolute Error (MAE) estimator

- Absolute error cost function

$$C[\theta, \hat{\theta}(x)] = C(\varepsilon) = |\varepsilon|$$

- The corresponding optimal estimator is *the median of the posterior distribution*

$$\int_{-\infty}^{\hat{\theta}} p(\theta|x)d\theta = \int_{\hat{\theta}}^{\infty} p(\theta|x)d\theta$$

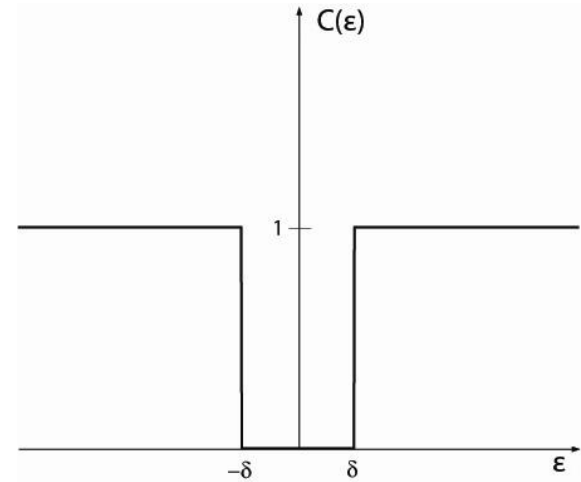


🔥 The Maximum a Posteriori (MAP) Estimator

- Hit-or-miss cost function

$$C(\varepsilon) = \begin{cases} 0, & |\theta - \hat{\theta}| < \delta \\ 1, & \text{otherwise} \end{cases}$$

- The corresponding optimal estimator is the *mode of the posterior distribution*



$$\hat{\theta}(x) = \operatorname{argmax}_{\theta} p(\theta|x) = \operatorname{argmax}_{\theta} p(x|\theta)p(\theta)$$

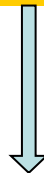
🔥 Example (1)

- Assume that $p(x|\theta) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(x-\theta)^2}{2\sigma^2}\right)$

and that the prior pdf is $p(\theta) = \frac{\gamma}{\pi(\theta^2 + \gamma^2)}$

- The MAP estimator can be found as follows:-

$$\hat{\theta}(x) = \operatorname{argmax}_{\theta} [\ln p(x|\theta) + \ln p(\theta)]$$



$$\hat{\theta}(x) = \operatorname{argmax}_{\theta} \left[-\frac{(x-\theta)^2}{2\sigma^2} + \ln \frac{\gamma}{\pi(\theta^2 + \gamma^2)} \right]$$

🔥 Example (continued)

$$\hat{\theta}(x) = \operatorname{argmax}_{\theta} \left[-\frac{(x - \theta)^2}{2\sigma^2} + \ln \frac{\gamma}{\pi(\theta^2 + \gamma^2)} \right]$$

- Differentiating with respect to θ

$$\frac{d}{d\theta} \left[-\frac{(x - \theta)^2}{2\sigma^2} + \ln \frac{\gamma}{\pi(\theta^2 + \gamma^2)} \right] = \frac{x - \theta}{\sigma^2} - \frac{2\theta}{\theta^2 + \gamma^2}$$

- Setting equal to 0 yields

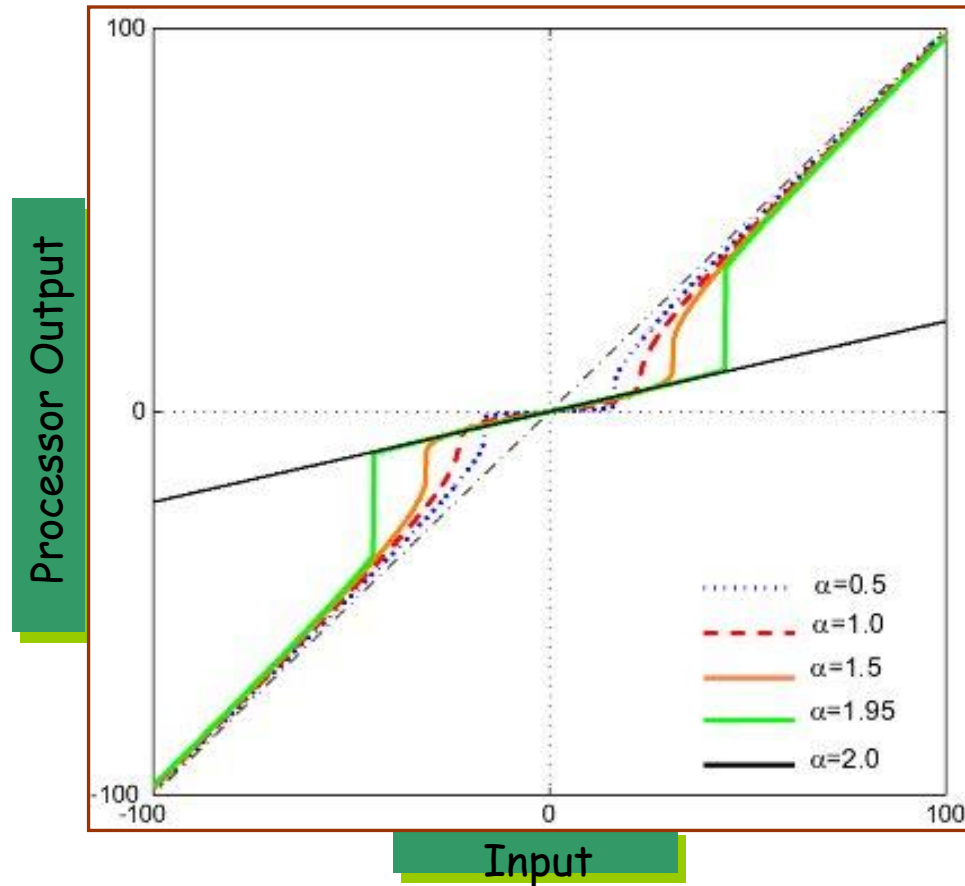
$$\frac{x - \theta}{\sigma^2} = \frac{2\theta}{\theta^2 + \gamma^2}$$

- Finally, rearranging

$$\theta^3 - x\theta^2 + (\gamma^2 + \sigma^2)\theta - \gamma^2x = 0$$



Example: MAP "Processor" I/O Curves



🔥 Example 2

- Assume that the likelihood function is exponential:

$$p(x|\theta) = \begin{cases} \theta \exp(-\theta x), & \text{for } x > 0 \\ 0, & \text{for } x < 0 \end{cases}$$

and that the prior pdf is

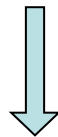
$$p(\theta) = \begin{cases} \lambda \exp(-\lambda\theta), & \text{for } \theta > 0 \\ 0, & \text{for } \theta < 0 \end{cases}$$

$$\hat{\theta}_{MAP} = ?$$

🔥 Example 2 - Exponential prior

- The MAP estimator can be found as follows:-

$$\hat{\theta}(x) = \operatorname{argmax} [\ln p(x|\theta) + \ln p(\theta)]$$



$$\begin{aligned}\hat{\theta}(x) &= \operatorname{argmax} [\ln[\theta \exp(-\theta x)] + \ln[\lambda \exp(-\lambda \theta)]] = \\ &= \operatorname{argmax} [\ln \theta - \theta x + \ln \lambda - \lambda \theta]\end{aligned}$$

- Differentiating with respect to θ

$$\frac{d}{d\theta} [\ln \theta - \theta x + \ln \lambda - \lambda \theta] = \frac{1}{\theta} - x - \lambda$$

- Setting equal to 0 yields

$$\hat{\theta} = \frac{1}{x + \lambda}$$

🔥 Example 3 - Gaussian prior

- Assume that the likelihood function is

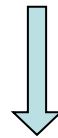
$$p(x|\theta) = \frac{1}{\sigma_n \sqrt{2\pi}} \exp - \frac{(x - \theta)^2}{2\sigma_n^2}$$

and that the prior pdf is

$$p(\theta) = \frac{1}{\sigma \sqrt{2\pi}} \exp - \frac{\theta^2}{2\sigma^2}$$

- The MAP estimator can be found as follows:-

$$\hat{\theta}(x) = \operatorname{argmax} [\ln p(x|\theta) + \ln p(\theta)]$$



$$\hat{\theta}(x) = \operatorname{argmax} \left[-\frac{(x - \theta)^2}{2\sigma_n^2} - \frac{\theta^2}{2\sigma^2} \right]$$

🔥 Example 3 (continued)

$$\hat{\theta}(x) = \operatorname{argmax} \left[-\frac{(x - \theta)^2}{2\sigma_n^2} - \frac{\theta^2}{2\sigma^2} \right]$$

- Differentiating with respect to θ

$$\frac{d}{d\theta} \left[-\frac{(x - \theta)^2}{2\sigma_n^2} - \frac{\theta^2}{2\sigma^2} \right] = \frac{x - \theta}{\sigma_n^2} - \frac{\theta}{\sigma^2}$$

- Setting equal to 0 yields

$$\frac{x - \theta}{\sigma_n^2} = \frac{\theta}{\sigma^2}$$

- Finally:-

$$\hat{\theta} = \frac{\sigma^2}{\sigma^2 + \sigma_n^2} x$$

🔥 Example 4: Laplace Prior - soft thresholding

- Assume that the likelihood function is

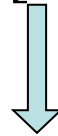
$$p(x|\theta) = \frac{1}{\sigma_n \sqrt{2\pi}} \exp - \frac{(x - \theta)^2}{2\sigma_n^2}$$

and that the prior pdf is

$$p(\theta) = \frac{1}{\sigma \sqrt{2}} \exp - \frac{\sqrt{2}|\theta|}{\sigma}$$

- The MAP estimator can be found as follows:-

$$\hat{\theta}(x) = \operatorname{argmax} [\ln p(x|\theta) + \ln p(\theta)]$$



$$\hat{\theta}(x) = \operatorname{argmax} \left[-\frac{(x - \theta)^2}{2\sigma_n^2} - \frac{\sqrt{2}|\theta|}{\sigma} \right]$$

🔥 Example 4: Laplace Prior - soft thresholding

$$\hat{\theta}(x) = \operatorname{argmax} \left[-\frac{(x - \theta)^2}{2\sigma_n^2} - \frac{\sqrt{2}|\theta|}{\sigma} \right]$$

- Differentiating with respect to θ

$$\frac{d}{d\theta} \left[-\frac{(x - \theta)^2}{2\sigma_n^2} - \frac{\sqrt{2}|\theta|}{\sigma} \right] = \frac{x - \theta}{\sigma_n^2} - \frac{\sqrt{2}}{\sigma} \operatorname{sign}(\theta)$$

- Setting equal to 0 yields

$$\hat{\theta} = x - \sqrt{2} \frac{\sigma_n^2}{\sigma} \operatorname{sign}(\theta)$$

- Finally:-

$$\hat{\theta} = \operatorname{sign}(x) \left(|x| - \sqrt{2} \frac{\sigma_n^2}{\sigma} \right)_+, \text{ where } (g)_+ = \begin{cases} 0, & g < 0 \\ g, & \text{otherwise} \end{cases}$$

🔥 Soft thresholding and other Bayesian Estimators

- Behaviour comparison for four different estimators
 - Cauchy thresholding (CT)
 - Soft thresholding (ST)
 - Amplitude scale invariant Bayes estimator (ABE)

